WamBANet: A Wavelet-Guided State-Space Modeling and Semantic Regularization for Weakly Supervised Salient Object Detection in Optical Remote Sensing Images

Tae Hun Lee, Seunghyeon Yu, and Jae Yeol Lee, *Member, IEEE*

***Abstract*—Salient object detection (SOD) in optical remote sensing images (ORSIs) remains fundamentally challenging due to extreme object-scale variations, complex background textures, and ambiguous foreground–background separations. While scribble-based weak supervision significantly reduces annotation costs, the inherent sparsity of scribble labels introduces significant structural uncertainty during dense prediction, frequently resulting in boundary erosion and interior collapse. Existing approaches alleviate this issue by incorporating external edge-detection networks or boundary pseudo-labels, but such strategies incur additional computational overhead and often introduce noisy background edges that degrade detection accuracy. To address these limitations, we present WamBANet, a wavelet-guided state-space modeling framework for scribble-supervised ORSI-SOD that enhances structural boundary cues without external edge detectors or boundary pseudo-labels. The proposed wavelet state-space model boundary attention (WamBA) module couples wavelet-derived high-frequency structural priors with SS2D-based low-frequency contextual modeling, enabling complementary semantic–structural refinement under sparse supervision. In addition, the semantic-guided attention enhancement (SGAE) module injects high-level semantic priors into low-level representations to reinforce interior–boundary consistency and suppress hollow-region artifacts. Extensive experiments on the ORSSD and EORSSD datasets demonstrate that WamBANet consistently outperforms state-of-the-art (SOTA) weakly supervised methods and achieves performance comparable to fully supervised models. Cross-dataset evaluations and camouflage object detection experiments further demonstrate the generalization and extensibility of the proposed WamBANet model. Code and experimental results will be made publicly accessible at https://github.com/HAID-LAB/WamBANet**

***Index Terms*—Optical remote sensing image (ORSI), scribble, wavelet transform, weakly supervised salient object detection.**

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I. Introduction

SALIENT object detection (SOD) aims to identify visually distinctive objects or regions within an image. SOD serves as a critical preliminary stage for a diverse array of downstream computer vision tasks, encompassing image classification [1], [2], semantic segmentation [3], [4], [5], [6], object detection [7], and change detection [8]. While natural scene image SOD (NSI-SOD) has advanced rapidly with the development of deep learning paradigms, ORSI-SOD continues to present substantially greater algorithmic challenges. Remote sensing imagery, typically acquired by satellites or drones, often exhibits cluttered backgrounds, ambiguous foreground–background separations, extreme object-scale variations, weather conditions, and multiple spatially distributed objects (e.g., ships, buildings, and aircraft). Consequently, these unique geospatial characteristics significantly amplify topological complexity, fundamentally distinguishing the challenges of ORSI-SOD from those of its natural scene counterpart [1], [7], [9].

Most existing SOD approaches are fully supervised and rely on precise pixel-level annotations. While such methods achieve strong performance, generating dense annotations for high-resolution remote sensing imagery is extremely costly and labor-intensive. To alleviate this burden, weakly supervised learning, which leverages incomplete annotations such as points, bounding boxes, and scribbles, has gained increasing attention. As illustrated in Fig. 1, weak supervision aims to enable efficient model training using simplified annotation forms rather than full pixel-level ground truth. Among these, scribble annotations offer a favorable tradeoff between labeling efficiency and structural guidance, providing coarse object location and shape priors with minimal human effort.

Because of complex backgrounds and diverse object scales, accurately identifying object boundaries in ORSI is inherently challenging. Some fully supervised methods attempt to mitigate this issue by incorporating auxiliary boundary annotations into the training pipeline [10]. However, incomplete annotations provide only coarse object information and lack precise boundary details, making accurate boundary localization difficult. In particular, the propagation of sparse scribble labels to

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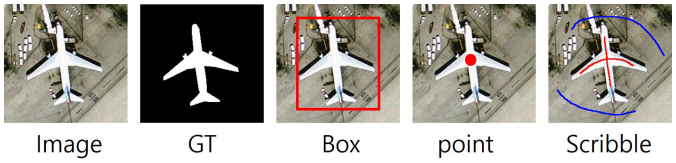


Fig. 1. Image- and pixel-level ground truths along with incomplete annotations widely used in SOD, such as bounding boxes, points, and scribbles. Red scribbles represent the background, and blue scribbles indicate the salient object.

dense pixel-level predictions inherently involves severe structural uncertainty, frequently manifesting as boundary collapse or incomplete predictions within object interiors. Therefore, some weakly supervised methods [11], [12] adopt auxiliary boundary detection networks (e.g., richer convolutional features (RCFs) [13] and holistically nested edge detection (HED) [14]) to precompute boundaries and incorporate them during training. Furthermore, some weakly supervised methods [11], [15] employ Canny edge detection [16] to derive boundary cues from feature maps.

However, these approaches involve extra CPU–GPU data transfer and precomputation, reducing computational efficiency. They also require additional networks or preprocessing for boundary extraction, increasing overall model complexity and cost. Moreover, boundaries extracted via conventional methods frequently encompass edges originating from non-salient background entities; without rigorous filtering mechanisms, these irrelevant structural cues inevitably degrade comprehensive model performance.

From a modeling perspective, ORSI-SOD requires effective long-range contextual reasoning due to large spatial extents and heterogeneous scene composition. Transformer-based architectures address this by modeling global self-attention, but their quadratic computational complexity limits scalability for high-resolution remote sensing imagery. Recently, selective state-space models (SSMs)—most notably Mamba-based architectures—have demonstrated strong capability for modeling long-range dependencies while maintaining linear computational complexity. Nevertheless, the reliance on sequential 1-D scanning mechanisms inherently disrupts the preservation of spatial adjacency when these models are directly applied to 2-D visual imagery, potentially causing structural distortion in complex ORSI scenes.

These observations highlight three key limitations of existing weakly supervised ORSI-SOD methods as follows.

1) Boundary enhancement often relies on auxiliary edge-detection networks or externally generated pseudo-labels.
2) Efficient global modeling remains challenging without compromising spatial structural coherence.
3) Frequency-domain structural priors have not been fully exploited to guide boundary-aware feature refinement under sparse supervision.

To address these challenges, we propose **WamBANet**, a frequency-guided state-space modeling framework for weakly supervised ORSI-SOD. By coupling wavelet-derived structural priors with SS2D-based contextual modeling, the proposed method refines long-range features while mitigating boundary collapse without relying on auxiliary edge-detection networks.

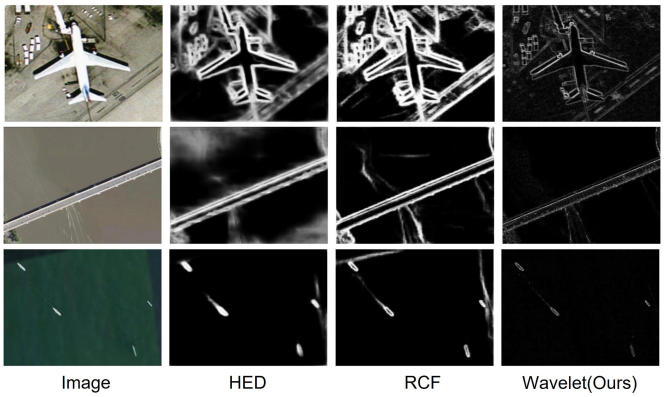


Fig. 2. Comparison of boundary extraction results obtained by different methods.

Specifically, WamBANet leverages a MobileViT encoder and introduces two key architectural components. First, the wavelet state-space model boundary attention (WamBA) module integrates multidirectional wavelet-based frequency decomposition with 2-D selective scan (SS2D) state-space modeling. This mechanism enhances boundary cues while promoting coherent global feature propagation. In particular, the WamBA module directly extracts frequency boundary information from the input image via DWT, eliminating the need for additional edge-detection networks. As illustrated in Fig. 2, conventional RCF- and HED-based methods often produce thick boundary predictions and include erroneous background edges. When such noisy boundaries are treated as ground truth during training, they may introduce misleading supervision signals unrelated to actual object contours. In contrast, the proposed method integrates wavelet-based structural priors with Mamba-based selective state-space modeling to selectively emphasize salient object boundaries while suppressing background interference.

Second, the semantic-guided attention enhancement (SGAE) module serves as a top-down semantic regularizer, injecting high-level contextual representations into low-level feature maps. By addressing the representational imbalance between boundary-rich shallow layers and semantically dense deep layers, SGAE enforces consistency between object interiors and boundaries. The SGAE module further enhances structural coherence by injecting object-level semantic cues into shallow feature maps. Low-level features contain rich boundary details but lack high-level semantic discrimination. This representational imbalance may weaken consistency between object interiors and boundaries. By transferring semantic priors from deeper layers to shallower layers, SGAE strengthens cross-scale semantic correlations and promotes more coherent object reconstruction under sparse supervision.

Unlike prior scribble-supervised ORSI-SOD methods that compensate for sparse labels by relying on external edge detectors, boundary pseudo-labels, or additional preprocessing pipelines, WamBANet reformulates boundary preservation as an internal frequency-guided feature modeling problem based

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on state-space modeling. The key novelty is not the independent use of wavelet transform or state-space modeling, but their task-specific coupling. Wavelet-derived high-frequency responses provide structural boundary priors, while SS2D models low-frequency semantic-contextual dependencies with linear complexity. This complementary modeling enables WamBANet to preserve object completeness and boundary fidelity under sparse scribble supervision without introducing external edge supervision or auxiliary boundary-generation networks.

The key contributions of this work are summarized as follows.

1) *Frequency-Guided State-Space Modeling for Scribble-Supervised ORSI-SOD*: We propose a weakly supervised ORSI-SOD framework that addresses structural uncertainty caused by sparse scribble annotations by internally deriving boundary-aware priors from wavelet high-frequency responses. Unlike previous edge-supervised methods that require external edge detectors or boundary pseudo-labels, WamBANet incorporates structural cues directly into the feature modeling process.
2) *Wavelet State-Space Model Boundary Attention*: We introduce the WamBA module, which couples an SS2D branch and a wavelet edge detector branch. The SS2D branch models low-frequency semantic and contextual dependencies, whereas the wavelet branch preserves high-frequency boundary structures. Their integration produces refined feature maps that improve both object completeness and boundary fidelity under sparse supervision.
3) *Semantic-Guided Attention Enhancement*: We design the SGAE module to inject high-level object semantics into low-level feature representations. This top-down semantic regularization alleviates hollow interiors and foreground–background ambiguity, which frequently occur when dense saliency maps are learned from sparse scribble labels.
4) *Comprehensive Empirical Validation and Generalization Analysis*: We conduct extensive quantitative and qualitative evaluations on ORSSD and EORSSD, benchmarking against 18 state-of-the-art (SOTA) methods. The proposed model not only outperforms existing weakly supervised approaches across all six evaluation metrics but also achieves performance competitive with fully supervised approaches. Cross-dataset validation and additional camouflage object detection experiments further confirm the extensibility and generalization of the proposed WamBANet.

II. Related Work

ORSI-SOD is recognized as a more challenging task than general SOD due to extreme object scale variations in high-resolution scenes, complex background textures, and low-contrast or ambiguous boundaries [17], [18], [19], [20], [21], [22]. Previous studies on ORSI-SOD are categorized into three groups: 1) fully supervised models and their limitations; 2) structural uncertainty in weakly supervised learning; and 3) the emerging trend of state-space models (SSMs, including Mamba). Based on this review, we discuss how the proposed SSM-based boundary/context enhancement model bridges existing gaps.

A. Fully Supervised ORSI-SOD

Early deep-learning-based ORSI-SOD methods were primarily developed under fully supervised settings with precise pixel-level annotations. For example, LVNet [17] employed a nested encoder–decoder architecture with a dual-stream pyramid to handle multiscale objects, while DAFNet [18] densely propagated shallow-layer attention information to deeper layers to alleviate spatial information loss. In addition, joint modeling of boundary and region information was proposed [19]. Furthermore, cross-scale interaction with enlarged receptive fields was explored to strengthen contextual representation [20].

More recently, extensive efforts have focused on leveraging the global self-attention mechanism of Transformers to directly model long-range dependencies [23], [24], [25]. For instance, ASNet [25] combined CNN and Transformer architectures to simultaneously capture local details and global context. Adaptive spatial tokenization has also been proposed to improve token efficiency for high-resolution ORSI [24]. In addition, semantic-guided attention refinement has been explored to improve feature discrimination [26].

In addition to these developments, recent studies have improved ORSI-SOD performance from complementary perspectives. ERNet [27] introduced additional boundary annotations during training to calibrate uncertain information around object boundaries. LGIPNet [28] refined salient object representations by integrating global and local information through a dual-domain attention module. SESRNet [29] exploited spatial–frequency information to effectively capture long-range contextual dependencies. DarkSalNet [30] enabled ORSI-SOD in low-light conditions by leveraging UVI representations.

However, fully supervised approaches incur extremely high annotation costs. Moreover, as networks become deeper and more complex to improve performance, the computational burden for both training and inference increases significantly [10], [31], [32], [33].

B. Scribble-Based Weakly Supervised ORSI-SOD

To reduce labeling costs, weakly supervised learning utilizing incomplete annotations such as image-level tags or scribbles has been actively studied. In particular, scribble annotations are considered a promising alternative due to their high labeling efficiency while still providing object location and overall shape. SBA-Net [11] introduced scribble supervision into ORSI-SOD and attempted to refine structure using boundary label generation based on class activation maps (CAMs) [34] along with a boundary-aware module. SHNet [12] performed hierarchical feature learning through scribble attention mechanisms.

Motivated by these advantages, scribble-based weakly supervised learning has recently been extended beyond ORSI-SOD to various remote sensing image analysis tasks.

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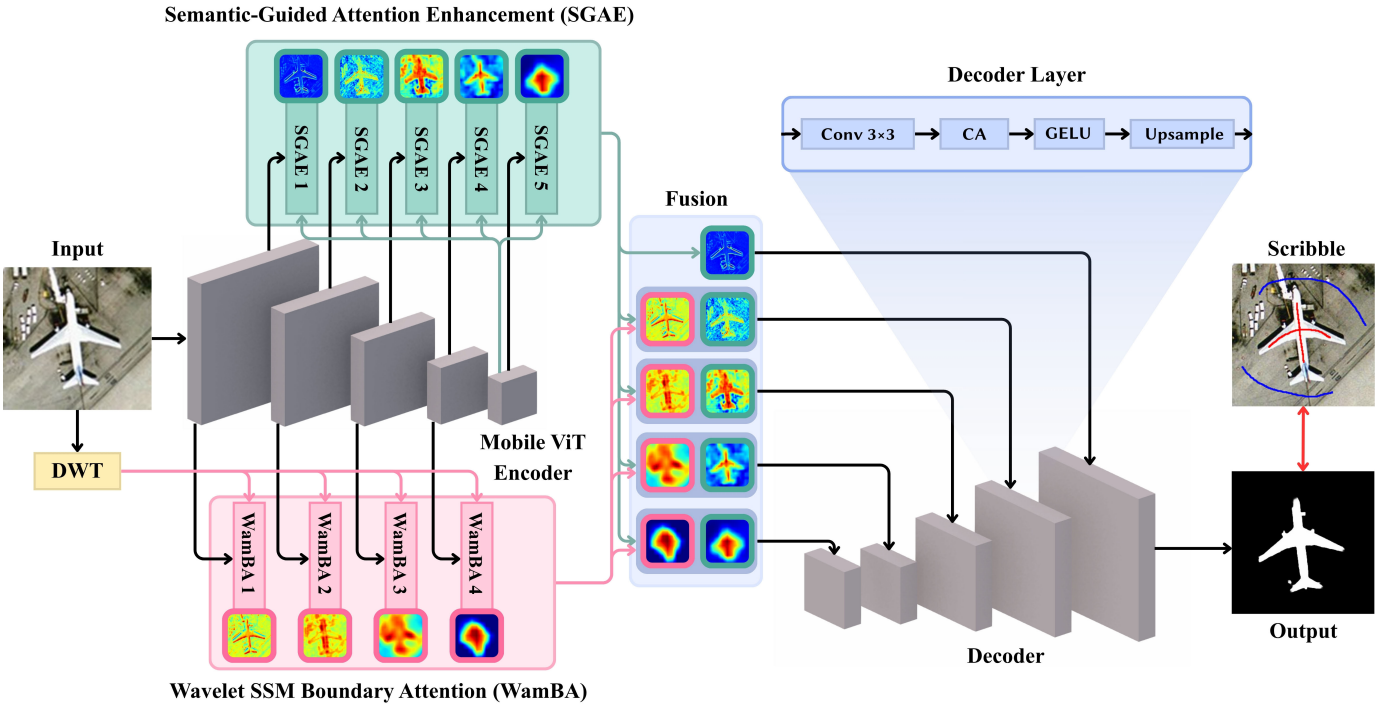


Fig. 3. Network architecture of WamBANet.

For example, scribble annotations have been utilized in road extraction [35], semantic segmentation [36], and multimodal change detection [37] to enable fine-grained pixel-level prediction with limited supervision. These studies indicate that scribble annotations can provide positional and structural cues for objects or changed regions at a substantially reduced labeling cost, thereby demonstrating their applicability to diverse remote sensing image analysis tasks.

Nevertheless, weak supervision inherently involves structural uncertainty during the propagation of sparse labels to entire object regions, often resulting in collapsed boundaries or hollow interior predictions [11], [12]. Thus, most of the previous studies have typically introduced auxiliary networks, postprocessing strategies, or complex constraints, which conflict with model simplicity and efficiency. Therefore, there is a growing need for approaches that can simultaneously strengthen boundary cues and global context understanding without introducing additional architectural complexity.

C. SSM-Based Vision Models

Although Transformer architectures are powerful, they typically incur a computational complexity of $O(N^2)$ with respect to the token sequence length, which becomes computationally expensive for high-resolution ORSI [23]. Recently, selective SSMs, represented by Mamba [21], have attracted attention due to their ability to model long-range dependencies with linear complexity ($O(N)$), making them suitable for high-resolution vision tasks. Extensions to visual backbones such as VMamba [22], hybrid architectures including MambaVision [38] and Samba [39], as well as applications to remote sensing segmentation [3], have been reported. Additionally, structured SSM research underlying SSMs has continuously advanced from the perspective of long-sequence modeling [40], [41].

However, converting 2-D images into 1-D sequences during scanning may weaken spatial adjacency preservation, potentially leading to contextual distortion in ORSI scenes with complex topological structures [22]. Therefore, in ORSI-SOD, it is crucial to combine 2-D scanning/refinement strategies that preserve spatial structure with enhanced boundary cues.

In summary, weakly supervised approaches such as scribbles often require heavy architectural designs to mitigate structural collapse [11], [12]. Transformer-based global modeling incurs increasing computational overhead in high-resolution scenarios [23], while SSM/Mamba offers improved efficiency but still faces challenges in preserving spatial adjacency [21], [22].

III. Proposed Method

A. Overall Architecture

This section presents an overview of the proposed WamBANet model to address the challenges of existing weakly supervised approaches in ORSI-SOD. WamBANet aims to simultaneously achieve high accuracy and computational efficiency without introducing auxiliary edge-detectors or complex postprocessing.

Fig. 3 illustrates the overall network architecture of the proposed model. First, multiscale feature maps extracted from the encoder MobileViT [31] are fed into the WamBA module, where boundary information is enhanced through the wavelet transform and Mamba-based state-space modeling. Simultaneously, the SGAE module injects high-level feature maps into low-level feature maps, compensating for the limitation of

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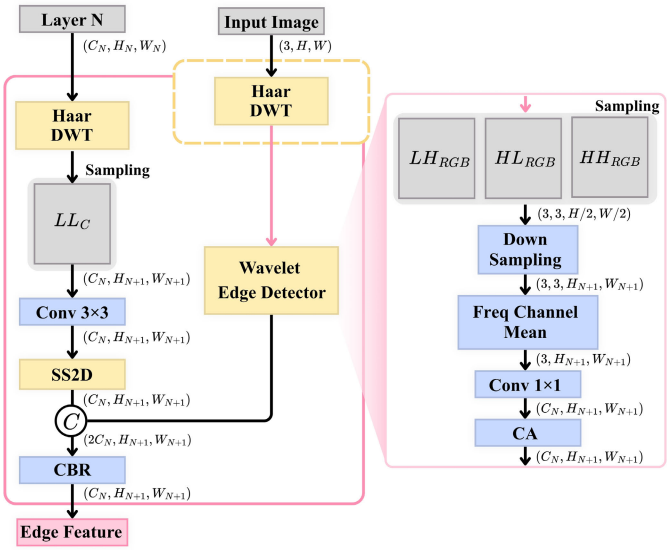


Fig. 4. WamBA module.

low-level features that contain rich boundary cues but insufficient semantic information. Finally, the decoder integrates the refined feature maps and progressively restores spatial resolution, thereby generating a saliency map that includes accurate object localization and refined boundaries even under scribble-based weak supervision.

B. WamBA Module for Boundary Awareness

To achieve boundary awareness without external dependencies, we introduce the WamBA module as shown in Fig. 4. The WamBA module consists of two key components: the wavelet edge detector branch and the SS2D branch. WamBA extracts high-frequency boundary cues from the initial input image I using DWT and combines them with multistage feature maps from the encoder to obtain boundary representations that are spatially precise and consistent. This design suppresses boundary interference from background and nonsalient objects while emphasizing structural contours of salient objects. Formally, the initial image I is decomposed into low–high, high–low, and high–high (LH, HL, and HH) frequency subbands through DWT

$$\{LH, HL, HH\} = \text{Wavelet}(I). \tag{1}$$

To aggregate multidirectional boundary information, the subbands are averaged and subsequently refined via a 3×3 convolution and channel attention (CA) to yield a highly discriminative high-frequency feature representation $\hat{T}$

$$\hat{T} = \text{CA}(\text{Conv}(\text{Mean}(\text{Down}(LH, HL, HH)))). \tag{2}$$

Concurrently, DWT is applied to the multiscale features E^i extracted from the encoder to isolate the low-frequency structural components LL^i. To effectively establish global receptive fields and capture long-range dependencies across the ORSI scene without disrupting spatial coherence, the low-frequency features are processed through an SS2D SSM [22]. Unlike traditional self-attention mechanisms that scale quadratically with spatial resolution, SS2D achieves linear computational complexity while facilitating omnidirectional contextual modeling

$$\{LL^i\} = \text{Wavelet}(E^i), \quad i \in \{1, 2, 3\} \tag{3}$$

$$L^i = \text{Conv}(LL^i) \tag{4}$$

$$\hat{L}^i = \text{SS2D}(Norm(L^i)) + L^i, \quad i \in \{1, 2, 3\}. \tag{5}$$

Finally, the refined low-frequency information $\hat{L}^i$ and high-frequency information $\hat{T}$ are concatenated and further refined by convolution-BatchNorm-ReLU (CBR) to generate the boundary-enhanced feature map M^i

$$M^i = \text{CBR}(\text{Concat}(\hat{T}, \hat{L}^i)), \quad i \in \{1, 2, 3\}. \tag{6}$$

It should be noted that WamBA is not designed as a simple auxiliary edge branch. The SS2D branch and wavelet edge detector branch play complementary roles in addressing different failure modes of scribble supervision. The SS2D branch enhances low-frequency semantic and contextual responses, but it remains dependent on encoder features and may be unreliable when object evidence is weak or spatially ambiguous. In contrast, the wavelet branch extracts high-frequency structural responses directly from the input image, preserving boundary cues that are often weakened during progressive downsampling. By fusing these two representations, WamBA jointly refines semantic object responses and structural boundary details, thereby mitigating both interior collapse and boundary erosion.

The resulting feature map M^i has half the spatial resolution of the corresponding encoder feature map E^i and is used for subsequent feature fusion. This design guides state-space feature refinement using high-frequency wavelet-derived structural priors, yielding sharp and accurate object delineations under scribble-based weak supervision.

C. SGAE Module

Although the WamBA module strengthens boundary information within each feature map, boundary cues alone are insufficient for fully representing the salient object. In particular, low-level feature maps lack sufficient semantic information to describe the overall object structure. Therefore, merely enhancing boundaries cannot guarantee complete object representation. To address this issue, we propose a semantic injection strategy that integrates object-centric semantic information extracted from high-level feature maps into lower-level feature maps. As shown in Fig. 5, salient object information is extracted from the high-level feature map E^5 and injected into each feature level for enhancement. The corresponding operations are formulated as follows:

$$E^{attn} = \sigma(\text{Conv}_{1\times1}(E^5)) \tag{7}$$

$$E^i_{attn} = \begin{cases} \text{Up}(E^{attn}) \times E^i, & i \in \{1, 2, 3, 4\} \\ E^{attn} \times E^i, & i = 5 \end{cases} \tag{8}$$

$$E^i_c = \text{ASPP}(E^i_{attn}) + E^i. \tag{9}$$

The SGAE module extracts semantic representations as defined in (7)–(9). The feature map E^5 is reduced in channel dimensionality via a 1×1 convolution, followed by a sigmoid

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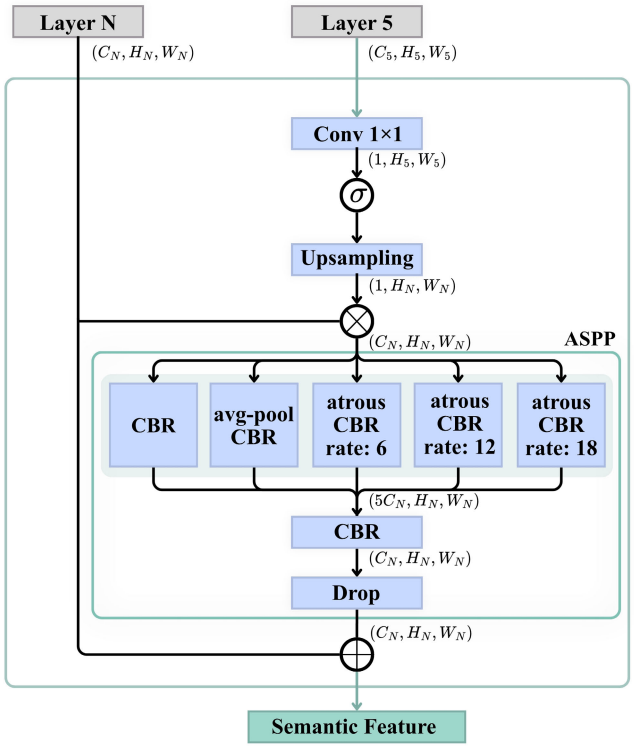


Fig. 5. SGAE module.

function to obtain the refined semantic attention map E^{attn}. The attention map is upsampled and multiplied with lower-level feature maps. Subsequently, atrous spatial pyramid pooling (ASPP) [42] is applied to integrate multilevel features through filters with diverse receptive fields. This process enables the generation of rich semantic representations that simultaneously reflect object structural characteristics and surrounding background context.

D. Decoder

As depicted in Fig. 3, the features generated by the WamBA and SGAE modules are concatenated through the channel dimension and then passed through the decoder to generate the prediction map. The decoder follows the sequence $\text{Conv}_{3\times3} \rightarrow$ CA $\rightarrow$ GeLU $\rightarrow$ Upsample.

After channel concatenation, $\text{Conv}_{3\times3}$ is applied to reduce the channel dimensionality. CA is then used to recalibrate interchannel importance. The refined feature map is subsequently processed by the GeLU activation function, upsampled, and fused with the higher-resolution feature map at the next stage. By repeating this procedure, spatial resolution is progressively restored to produce the final saliency prediction.

E. Loss Function

During training, optimization is performed by combining binary cross-entropy (BCE) and intersection-over-union (IoU) losses, which are widely used in ORSI-SOD. The prediction map P generated by WamBANet is trained to minimize pixel-level discrepancies with the scribble-based ground truth (GT), thereby simultaneously improving local classification accuracy and global region consistency. Each loss function is defined as follows:

$$\mathcal{L}_{BCE} = -\frac{1}{HW}\sum_{u,v}\Big[GT(u,v)\log P(u,v) + (1-GT(u,v))\log(1-P(u,v))\Big] \tag{10}$$

$$\mathcal{L}_{IoU} = 1 - \frac{\sum_{u,v} P(u,v)\,GT(u,v)}{\sum_{u,v}[P(u,v)+GT(u,v)-P(u,v)\,GT(u,v)]} \tag{11}$$

$$\mathcal{L}_{total} = \mathcal{L}_{BCE} + \mathcal{L}_{IoU}. \tag{12}$$

IV. Experiment

A. Datasets and Implementation Details

We conducted comparative experiments with 18 SOTA methods to evaluate the performance of WamBANet against existing approaches. Performance evaluation was conducted using two benchmark datasets, EORSSD [18] and ORSSD [17]. ORSSD consists of 800 images in total, including 600 training images and 200 testing images, while EORSSD is an extended version of ORSSD and contains 1400 training images and 600 testing images.

All experiments were performed in the PyTorch framework and conducted on an NVIDIA GeForce RTX 4090 GPU with 24 GB of memory. For training, we used the scribble annotations provided by SBA-Net [11]. All images were resized to 352 $\times$ 352 pixels, and vertical and horizontal flippings were applied to prevent overfitting. WamBANet was trained for 60 epochs with a batch size of 8. The starting learning rate was set to $1e{-}4$ and was reduced to one-tenth of the starting value after epoch 40.

B. Evaluation Metrics

For performance evaluation, we adopted six metrics widely used in existing SOTA methods: S-measure [43], (maximum, mean) F-measure [44], (maximum, mean) E-measure [45], and mean absolute error (MAE) [46].

The MAE measures the average absolute difference between predictions and ground truth, defined as follows:

$$\text{MAE} = \frac{1}{H\times W}\sum_{i=1}^{H}\sum_{j=1}^{W}\left|P(i,j) - G(i,j)\right| \tag{13}$$

where P denotes the prediction, G denotes the ground truth, and H and W represent the height and width of the image, respectively.

F-measure is computed based on precision and recall

$$F_\beta = \frac{\left(1+\beta^2\right)\times \text{Precision}\times \text{Recall}}{\beta^2\times \text{Precision} + \text{Recall}} \tag{14}$$

where β is set to 0.3.

E-measure jointly considers precision and alignment information between the predicted saliency map and GT, formulated as follows:

$$E_\phi = \frac{1}{H\times W}\sum_{i=1}^{H}\sum_{j=1}^{W}\phi\left(P_{i,j}, G_{i,j}\right) \tag{15}$$

where ϕ denotes the enhanced alignment matrix, and H and W represent the image height and width, respectively.

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TABLE I

COMPARATIVE EVALUATION RESULTS ON ORSI-SOD DATASETS USING S-MEASURE, (MAXIMUM, MEAN) F-MEASURE, (MAXIMUM, MEAN) E-MEASURE, AND MAE AS METRICS. ↑ INDICATES THAT HIGHER IS BETTER, AND ↓ INDICATES THAT LOWER IS BETTER. SUP. DENOTES THE SUPERVISION TYPE SUCH AS F (FULLY SUPERVISED LEARNING), S (SCRIBBLE), AND E (EDGE). THE TOP TWO RESULTS ARE HIGHLIGHTED IN RED BOLD AND BLUE UNDERLINE

Models	Backbone	Params (M) ↓	GFLOPs (G) ↓	Sup.	EORSSD [18]						ORSSD [17]					
					$S_\alpha \uparrow$	$F_\beta^{max} \uparrow$	$F_\beta^{mean} \uparrow$	$E_\xi^{max} \uparrow$	$E_\xi^{mean} \uparrow$	$M \downarrow$	$S_\alpha \uparrow$	$F_\beta^{max} \uparrow$	$F_\beta^{mean} \uparrow$	$E_\xi^{max} \uparrow$	$E_\xi^{mean} \uparrow$	$M \downarrow$
Fully Supervised Learning for ORSI																
DAFNet	VGG16	29.35	68.51	F	0.917	0.861	0.785	0.986	0.929	0.006	0.919	0.893	0.851	0.977	0.954	0.011
SARNet	Res2Net-50	40.30	75.1	F	0.929	0.877	0.857	0.967	0.961	0.009	0.935	0.906	0.887	0.971	0.966	0.010
CorrNet	VGG16	4.09	21.1	F	0.929	0.888	0.862	0.970	0.965	0.008	0.938	0.891	0.900	0.979	0.975	0.010
ASNet	PVT	65.43	20.1	F	0.935	0.896	0.877	0.978	0.975	0.006	0.944	0.917	0.907	0.980	0.976	0.008
PEFFNet	Swin	107.54	126.31	F	0.939	0.894	0.871	0.983	0.975	0.007	0.937	0.927	0.914	0.990	0.986	0.006
EANet	VGG16	25.2	30.1	F	0.929	0.897	-	0.974	-	0.007	0.931	0.918	-	0.975	-	0.011
RAMENet	MobileViT	5.18	8.72	F	0.942	0.897	0.884	0.982	0.978	0.005	0.949	0.923	0.915	0.986	0.982	0.007
LGIPNet	PVT	52.1	47.0	F	0.939	0.892	0.887	0.981	0.975	0.005	0.951	0.927	0.913	0.988	0.983	0.007
ERNet	PVT	67.1	58.78	F+E	0.943	0.898	0.884	0.984	0.979	0.005	0.953	0.929	0.915	0.987	0.983	0.007
SESRNet	SMT	22.38	38.92	F	0.946	0.900	0.888	0.983	0.980	0.004	0.955	0.932	0.921	0.989	0.984	0.006
Weakly Supervised Learning for ORSI																
WSSA	VGG16	15.53	70.6	S	0.852	0.778	0.771	0.925	0.901	0.015	0.848	0.814	0.806	0.908	0.908	0.027
SCWS	ResNet-50	63.54	53.9	S	0.654	0.721	0.626	0.836	0.737	0.094	0.725	0.801	0.722	0.886	0.799	0.096
SBA-Net	VGG16	21.2	371.9	S+E	0.861	0.816	0.798	0.938	0.922	0.014	0.863	0.837	0.825	0.939	0.919	0.026
SHNet	VGG16	20.71	110.5	S+E	0.868	0.784	0.761	0.923	0.915	0.018	0.856	0.791	0.779	0.915	0.909	0.034
PARF	ResNet-34	60.77	222.7	S	0.893	0.826	0.796	0.949	0.921	0.012	0.893	0.843	0.827	0.946	0.926	0.021
OMNet	ResNet-50	82.5	104.7	S	0.906	0.852	0.835	0.952	0.935	0.009	0.902	0.867	0.859	0.958	0.939	0.015
MIFA-Net	VGG16	83.5	102.6	S	0.905	**0.858**	**0.840**	0.960	0.935	0.008	0.918	0.888	0.880	0.969	0.962	0.013
LightEMNet	MobileNet	**6.60**	77.16	S+E	0.887	0.847	0.835	**0.962**	0.944	0.008	0.900	0.884	0.875	0.969	0.950	**0.012**
Ours	MobileViT	11.48	**25.22**	S	**0.915**	0.856	0.830	**0.962**	**0.953**	**0.007**	**0.926**	**0.892**	**0.881**	**0.970**	**0.963**	**0.012**

S-measure evaluates the structural similarity between the predicted saliency map and GT, defined as

$$S_\alpha = \alpha \cdot S_o + (1 - \alpha) \cdot S_r \tag{16}$$

where S_o denotes object-aware similarity, S_r denotes region-aware similarity, and α is set to 0.5.

C. Comparison With SOTA Methods

1) Quantitative Evaluation: We conducted comparative evaluations with 18 SOTA methods using widely adopted metrics, including S-measure, maximum and mean F-measure, maximum and mean E-measure, MAE, parameter count, and GFLOPs. The parameter count and GFLOPs were computed using the ptflops library [47].

Table I summarizes the comparative evaluation results. Among the compared approaches, ten are fully supervised models trained with complete pixel-level ground truth: DAFNet [18], SARNet [26], CorrNet [48], ASNet [25], PEFFNet [49], EANet [50], RAMENet [51], LGIPNet [28], ERNet [27], and SESRNet [29]. The remaining eight are scribble-based weakly supervised models, which are consistent with our setting: WSSA [52], SCWS [53], SBA-Net [11], SHNet [12], PARF [54], OMNet [55], MIFA-Net [56], and LightEMNet [15].

The quantitative results show that WamBANet outperforms previous weakly supervised ORSI-SOD methods. Specifically, on the EORSSD dataset, the proposed method achieves S_a of 0.915 (Ours) versus 0.906 (OMNet) and 0.887 (LightEMNet); E_ξ^{mean} of 0.953 (Ours) versus 0.944 (LightEMNet); and M of 0.007 (Ours) versus 0.008 (LightEMNet), demonstrating improvements across most metrics. Likewise, on the ORSSD dataset, the proposed method achieves S_a of 0.926 (Ours) versus 0.918 for MIFA-Net and 0.900 for LightEMNet. These results confirm that our model consistently achieves higher overall performance than existing weakly supervised methods.

In particular, compared with MIFA-Net, one of the most closely related prior approaches, the proposed method exhibits a substantially more lightweight architecture, requiring only approximately one-eighth of the parameters and one-fourth of the GFLOPs. In comparison with LightEMNet, although the proposed method involves a slightly larger number of parameters, it achieves considerably lower computational complexity with higher performance, reducing GFLOPs to nearly one-third of those of LightEMNet. Notably, LightEMNet depends on an external edge detector to generate auxiliary edge cues.

Furthermore, existing methods that exploit edge information as supervisory guidance [11], [12], [15] generally rely on additional edge extraction procedures based on external detectors such as HED or RCF. Although these operations are primarily performed during training, they still require separate preprocessing pipelines and additional edge-label generation processes, thereby increasing the overall complexity of the training framework.

In contrast, the proposed method directly leverages wavelet-derived high-frequency components to encode structural information, enabling effective incorporation of boundary-aware priors without requiring any additional edge-generation process. These results indicate that the proposed framework is advantageous not only in terms of segmentation performance, but also with respect to computational efficiency and the simplicity of the overall training pipeline.

As shown in Fig. 6, on both the EORSSD and ORSSD datasets, the PR curves of the proposed model are closest to

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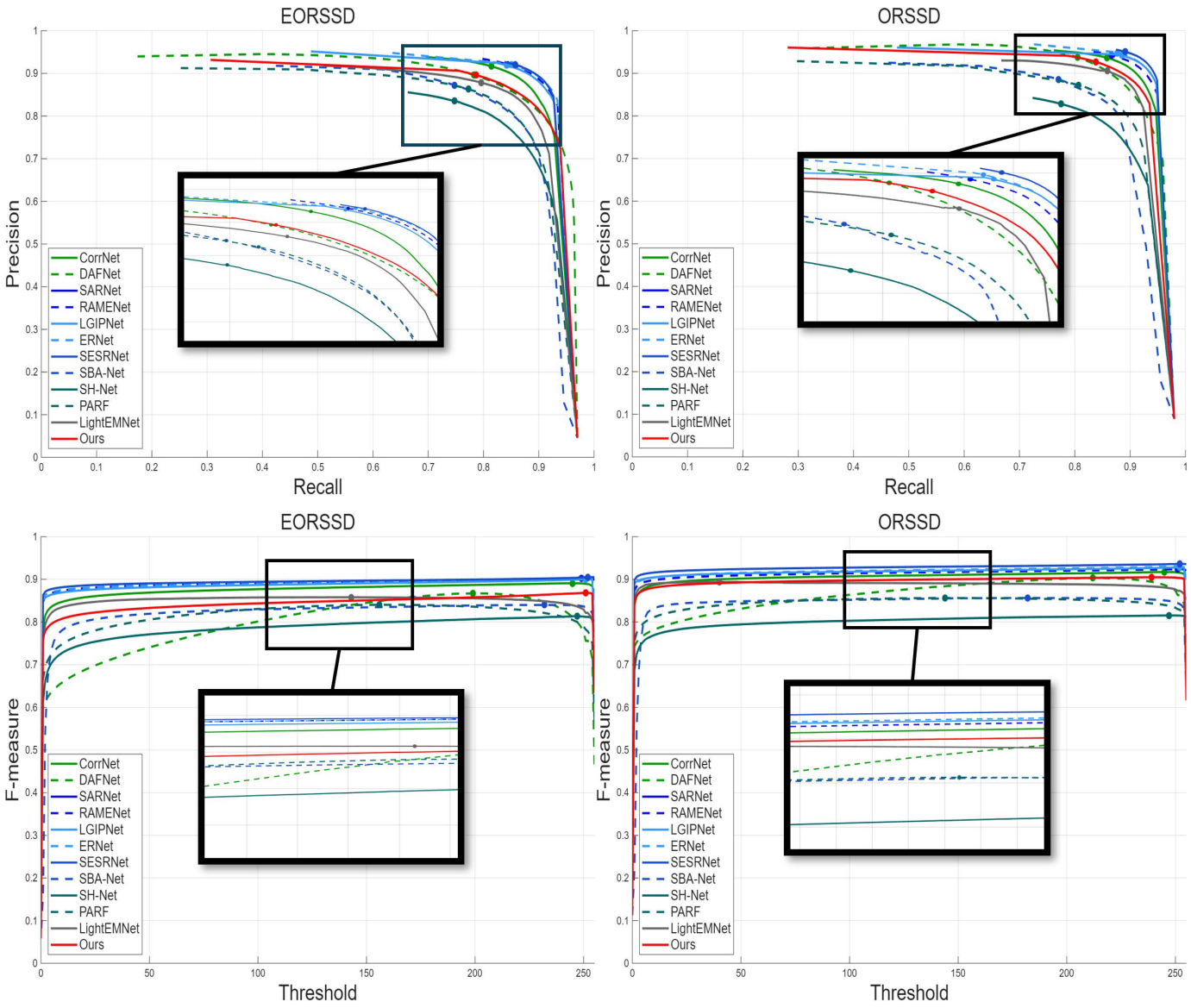


Fig. 6. PR curves and F-measure curves of the proposed model and compared SOTA models on two datasets.

the ideal point $(1, 1)$ among all weakly supervised methods. Notably, the proposed model also performs better than the fully supervised DAFNet in the PR-curve comparison. Similarly, in the F-measure curves, WamBANet achieves values closest to 1 among the compared weakly supervised methods. On the ORSSD dataset, it further achieves the highest F-measure among the compared weakly supervised methods. These results demonstrate that WamBANet achieves competitive performance not only against existing weakly supervised models but also against some fully supervised models.

2) Qualitative Evaluation: Fig. 7 shows comparative results on various complex and challenging images, including structurally complicated objects such as airplanes, ships, buildings, bridges, and rivers. We compared the proposed model with eleven SOTA models, including seven fully supervised models (CorrNet, DAFNet, SARNet, RAMENet, LGIPNet, ERNet, and SESRNet) and four weakly supervised models (SBA-Net, SHNet, PARF, and LightEMNet). To demonstrate the robustness of the proposed approach in ORSI-SOD, we conducted qualitative evaluation on five representative challenging object categories in the EORSSD dataset, as depicted in Fig. 7.

a) Airplane

Airplanes exhibit complex geometric structures in ORSIs, where extremities such as wings and tails frequently blend with background textures, complicating precise segmentation. In weakly supervised learning with scribbles, annotations for airplanes are not sufficiently precise, and in particular, incomplete scribbles for wings and tails limit the ability to learn the full object shape. As an example, in the first row of Fig. 7, all other weakly supervised models fail to clearly detect the tail region. In the second row, existing weakly supervised models also show unclear detection of wing regions. In contrast, the proposed model demonstrates superior segmentation performance by producing more consistent object shapes even in structurally complex regions such as tails and wings.

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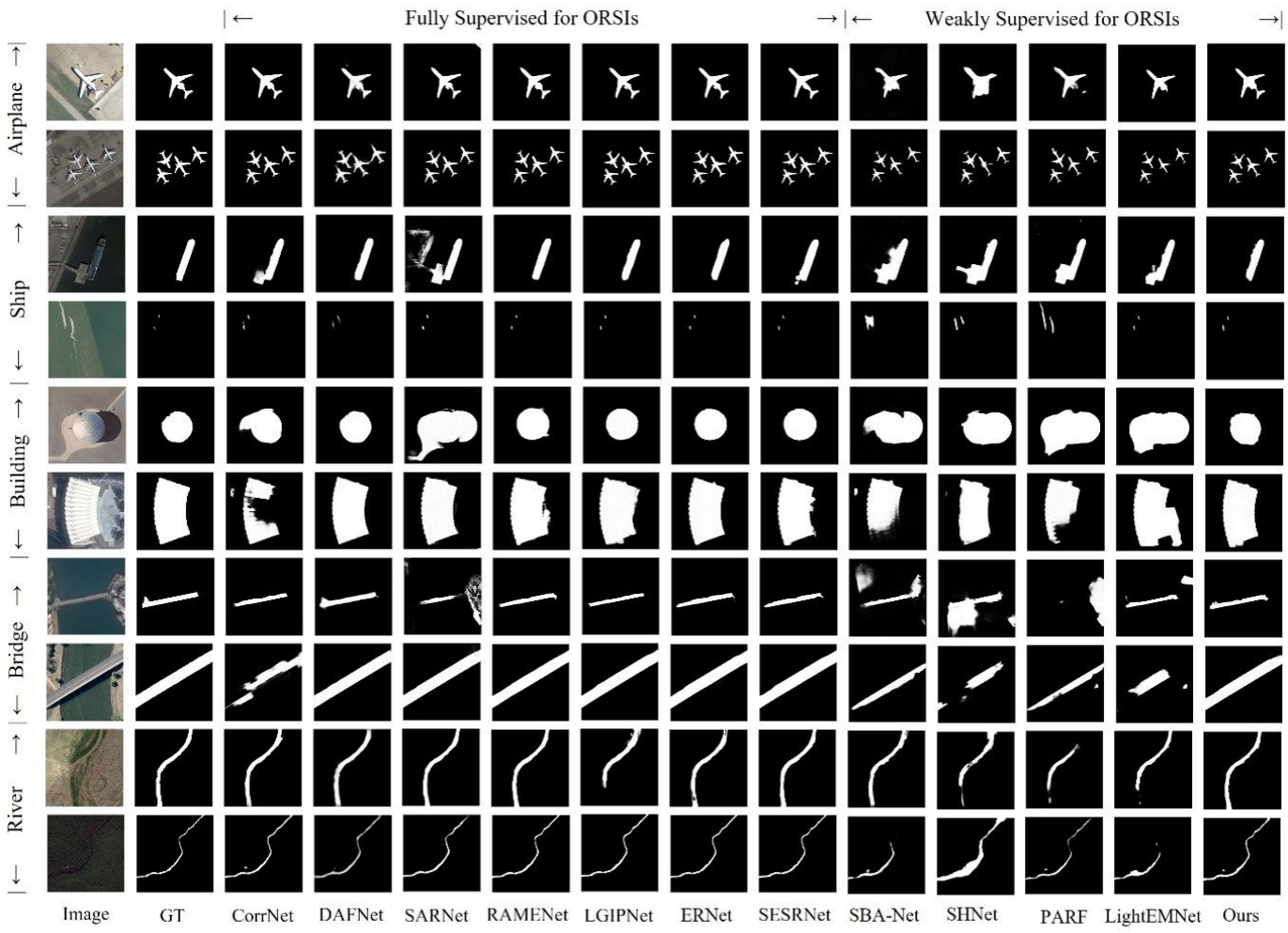


Fig. 7. Qualitative comparison between the proposed model and compared SOTA methods. Results are shown for airplanes, ships, buildings, bridges, and rivers.

b) Ship

Ships are also challenging objects in ORSI-SOD. They can be easily confused with the surrounding sea, leading to false detections. In addition, when ships are moving, wake trails may be mistakenly recognized as parts of the objects. In the third row of Fig. 7, existing weakly supervised models often produce false detections by confusing ships with surrounding structures. This issue appears in both weakly supervised methods and fully supervised models such as CorrNet and SARNet. Nevertheless, the proposed model achieves accurate detection under such conditions. Additionally, in the fourth row of Fig. 7, several weakly supervised models (e.g., SBA-Net, SHNet, and PARF) incorrectly include wake trails when detecting moving ships, whereas the proposed model successfully separates the wake from the ship, producing a more accurate segmentation result.

c) Building

Buildings are one of the most representative objects in ORSI, yet accurate detection remains difficult due to diverse shapes and complex backgrounds. In particular, shadows cast by buildings blur true object boundaries and make foreground–background separation more challenging. Moreover, the limitation of scribble annotations hinders learning sufficient information about full building regions. As shown in the fifth row of Fig. 7, existing weakly supervised models tend to produce inaccurate detections near building boundaries or over-detect nearby objects as buildings. In the sixth row, false detections occur in most weakly supervised models and even in some fully supervised models because the building roof color is similar to the surrounding environment. In contrast, the proposed model demonstrates robust detection performance not only at boundaries but also under confusing color conditions, producing more accurate segmentation results.

d) Bridge

Bridges are difficult to detect accurately due to their elongated structures and complex surrounding backgrounds. Adjacent regions such as water surfaces and roads often obscure boundaries, leading existing weakly supervised models to detect only partial bridge structures or to mistakenly include background regions. As shown in the seventh and eighth rows of Fig. 7, the proposed model effectively captures continuous structural characteristics and boundary cues of

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TABLE II

CROSS-VALIDATION TESTING ON THE EORSSD DATASET USING MODELS TRAINED WITH THE ORSSD DATASET

Models	ORSSD → EORSSD					
	$S_\alpha \uparrow$	$F_\beta^{\max} \uparrow$	$F_\beta^{\text{mean}} \uparrow$	$E_\xi^{\max} \uparrow$	$E_\xi^{mean} \uparrow$	$M \downarrow$
WSSA	0.765	0.627	0.645	0.809	0.847	0.026
SBA-Net	0.793	0.723	0.705	0.846	0.876	0.022
Ours	**0.874**	**0.784**	**0.766**	**0.918**	**0.908**	**0.014**

TABLE III

COMPARATIVE EVALUATION FOR CAMOUFLAGED OBJECT DETECTION USING SCRIBBLES. S DENOTES SCRIBBLE-BASED LEARNING METHODS, P DENOTES PSEUDO-ANNOTATION-BASED LEARNING METHODS USING SAM. F_β^w REPRESENTS THE WEIGHTED F-MEASURE

Models	Sup.	CAMO			
		$M \downarrow$	$S_\alpha \uparrow$	$E_\xi^{mean} \uparrow$	$F_\beta^w \uparrow$
WSSA	S	0.118	0.696	0.786	0.615
WS-SAM	S+P	0.092	0.759	0.818	0.667
CRNet	S	0.092	0.735	0.815	0.641
SEE	S+P	0.090	0.765	0.826	-
PPLNet	S	0.090	0.755	0.843	**0.697**
Ours	S	**0.088**	**0.772**	**0.852**	0.674

bridges, generating more complete and consistent detection results.

e) River

Rivers are large, irregularly shaped regions in ORSI and are often partially segmented or confused with surrounding terrain, resulting in false positives. In contrast, the proposed model effectively integrates global region structure and boundary information, producing more consistent and complete detection results. In the ninth row of Fig. 7, most weakly supervised models show false detections due to ambiguous boundaries caused by surrounding backgrounds, and in the tenth row, over-detection occurs due to confusion with nearby terrain. Despite these factors, the proposed model yields more accurate and stable segmentation results for river regions.

D. Cross-Validation and Generalization

We conducted cross-dataset validation to demonstrate the generalization and extensibility of the proposed model. First, the models trained on the ORSSD dataset were directly tested on the EORSSD dataset without training or fine-tuning. As shown in Table II, the proposed method outperforms previous approaches. Furthermore, we performed an additional experiment on a scribble-annotated camouflaged object detection dataset [57], as shown in Table III. For fair comparison, we evaluated WamBANet against representative weakly supervised SOD models and weakly supervised camouflaged object detection methods, including WS-SAM [58], CRNet [59], SEE [60], and PPLNet [61].

The experimental results show that the proposed model achieves superior performance compared to existing methods across most evaluation metrics, further demonstrating its effectiveness and generalization ability under weak supervision settings.

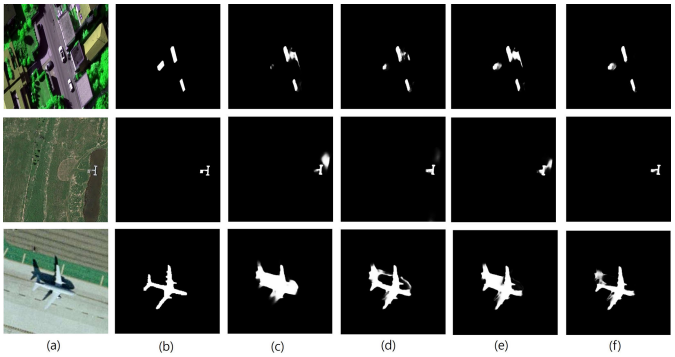


Fig. 8. Visualization of qualitative results concerning different module configurations of WamBANet. (a) Input images, (b) GT, (c) baseline, (d) baseline+WamBA, (e) baseline+SGAE, and (f) ours.

E. Ablation Study

1) Ablation Study of Different WamBANet Configurations: We performed ablation studies on the proposed model. For both datasets, we compared the performance impact of the WamBA module and the SGAE module. Table IV summarizes the results of the ablation study under four settings: 1) baseline; 2) w/WamBA; 3) w/SGAE; and 4) Ours. Baseline denotes an encoder–decoder architecture with both modules removed. As shown in Table IV, adding each module yields performance improvements across overall metrics.

Fig. 8 shows qualitative comparison results obtained by progressively adding the proposed modules, demonstrating that performance improves gradually compared with the baseline as each module is added. In particular, the result in the third row represents a challenging case where most ORSI-SOD models struggle to separate an airplane from its shadow. For the baseline model, the shadow region is detected together with the object, whereas in Baseline+SGAE, false detection of the shadow is effectively suppressed by utilizing semantic information. Moreover, Baseline+WamBA distinguishes the boundary between the object and the shadow more clearly through boundary-aware detection. Finally, the proposed WamBANet model successfully generates a more refined prediction map where the shadow and object are clearly separated by combining the advantages of both modules. These results demonstrate that the two modules complement each other and contribute to the final performance improvement.

2) Ablation Study on the WamBA Module: The WamBA module compensates for boundary information that tends to be lost as the model progressively reduces spatial resolution during feature extraction. Fig. 9 visualizes the behavior of the WamBA module shown in Fig. 4. Since the feature maps produced by WamBA have half the spatial resolution of the corresponding encoder features, they are spatially aligned with the feature maps at the next encoder stage. For instance, in A2, boundary information of the object becomes less distinct compared to A1, whereas in D1, the main contour of the object is preserved and the boundary with the background appears relatively clearer. The same tendency is observed when comparing A3 with D2 and A4 with D3. These observations suggest that WamBA stably preserves object boundary cues

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TABLE IV
ABLATION STUDY ON KEY MODULES OF WAMBANET

No.	Modules	EORSSD						ORSSD					
		$S_\alpha \uparrow$	$F_\beta^{\max} \uparrow$	$F_\beta^{\text{mean}} \uparrow$	$E_\xi^{\max} \uparrow$	$E_\xi^{mean} \uparrow$	$M \downarrow$	$S_\alpha \uparrow$	$F_\beta^{\max} \uparrow$	$F_\beta^{\text{mean}} \uparrow$	$E_\xi^{\max} \uparrow$	$E_\xi^{mean} \uparrow$	$M \downarrow$
1	Baseline	0.895	0.828	0.798	0.953	0.940	0.010	0.908	0.874	0.861	0.965	0.958	0.014
2	w/ WamBA	0.902	0.836	0.809	0.956	0.944	0.009	0.915	0.878	0.868	0.967	0.960	0.013
3	w/ SGAE	0.904	0.842	0.815	0.955	0.945	0.008	0.916	0.881	0.870	0.968	0.961	0.013
4	Ours	**0.915**	**0.856**	**0.830**	**0.962**	**0.953**	**0.007**	**0.926**	**0.892**	**0.881**	**0.970**	**0.963**	**0.012**

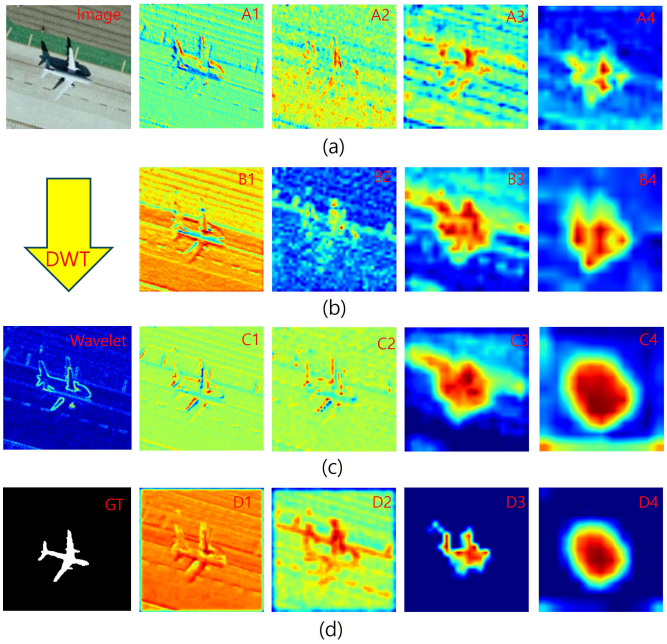


Fig. 9. Visualization of feature maps before, within, and after the WamBA module shown in Fig. 4. (a) A1–A4 show the input feature maps extracted from encoder stages 1–4, respectively. (b) B1–B4 show the corresponding outputs of the SS2D branch. (c) C1–C4 show the corresponding outputs of the Wavelet Edge Detector branch. (d) D1–D4 show the final refined feature maps produced by WamBA.

weakened by reduced spatial resolution and provides structural cues for subsequent feature fusion stages.

To further validate the effectiveness of the proposed WamBA module, we conducted an ablation study on its two core branches, namely the SS2D branch and the wavelet edge detector branch, to analyze the contribution of each branch to the overall model performance.

Fig. 9(b) visualizes the feature representations obtained after the low-frequency components extracted from the encoder feature maps are processed through the SS2D branch (see Fig. 4). As shown in B1–B4, the feature maps generated by the SS2D branch exhibit stronger responses concentrated on salient object regions compared with the original encoder features shown in A1–A4. This indicates that the SS2D branch effectively enhances long-range contextual modeling and semantic representation. However, because the SS2D branch fundamentally relies on the encoder features for information propagation, its performance may be limited when the original feature map contains ambiguous object information. For example, in A2, the salient object representation is relatively unclear, which consequently leads to limited object responses in B2.

Fig. 9(c) presents the feature maps generated after the high-frequency components obtained through wavelet decomposition are processed by the wavelet edge detector branch (see also Fig. 4). Since these high-frequency representations primarily encode object boundary information, they preserve structural characteristics relatively robustly even during progressive resolution reduction. For instance, while the object representation in A2 appears ambiguous, the corresponding feature map C2 retains substantially clearer boundary structures. This observation suggests that high-frequency information effectively compensates for the boundary and structural cues that are often degraded or lost in encoder features, thereby enabling more stable preservation of object morphology.

Subsequently, the low-frequency semantic features shown in B1–B4 and the high-frequency structural features shown in C1–C4 are combined to generate the refined feature representations shown in D1–D4. These results demonstrate that WamBA effectively exploits both low-frequency semantic information and high-frequency structural information to produce more discriminative feature representations. In particular, the refined feature map D2 shows that the relatively weak object response observed in B2 is effectively complemented by the structural boundary cues provided by C2, resulting in a more complete and clearer representation of the object structure.

These observations are further supported by the quantitative results presented in Table V. The table shows the ablation study results for the two branches of WamBA. Both the wavelet edge detector-only configuration (w/Wavelet Edge Detector) and the SS2D-only configuration (w/SS2D) achieve noticeable performance improvements over the baseline model. Furthermore, the full WamBA configuration (Ours), which simultaneously incorporates both components, achieves the best overall performance, demonstrating the complementary relationship between semantic contextual modeling and structural boundary enhancement.

Furthermore, to verify the effectiveness of SS2D applied in WamBA, we conducted an additional ablation study by replacing it with other modules: 1) Mamba; 2) self-attention; and 3) SS2D. As shown in Table VI, the Mamba setting yields lower performance than Self-Attention and SS2D, which suggests that direct 1-D sequence modeling without spatially structured scanning may weaken adjacency preservation. The Self-Attention setting achieves performance similar to SS2D in terms of E_{max}, but it shows lower performance in preserving

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TABLE V
ABLATION STUDY ON WAMBA BRANCHES

No.		EORSSD				ORSSD			
		$S_\alpha \uparrow$	$F_\beta^{\max} \uparrow$	$E_\xi^{\max} \uparrow$	$M \downarrow$	$S_\alpha \uparrow$	$F_\beta^{\max} \uparrow$	$E_\xi^{\max} \uparrow$	$M \downarrow$
1	Baseline	0.904	0.842	0.955	0.008	0.916	0.881	0.968	0.013
2	w/ Wavelet Edge Detector	0.912	0.850	0.960	0.008	0.923	0.888	0.969	0.013
3	w/ SS2D	0.909	0.847	0.957	0.008	0.920	0.886	0.968	0.013
4	Ours	**0.915**	**0.856**	**0.962**	**0.007**	**0.926**	**0.892**	**0.970**	**0.012**

TABLE VI
PERFORMANCE EVALUATION OF DIFFERENT LOW-FREQUENCY MODELING METHODS IN WAMBA

No.		EORSSD				ORSSD			
		$S_\alpha \uparrow$	$F_\beta^{\max} \uparrow$	$E_\xi^{\max} \uparrow$	$M \downarrow$	$S_\alpha \uparrow$	$F_\beta^{\max} \uparrow$	$E_\xi^{\max} \uparrow$	$M \downarrow$
1	Mamba	0.907	0.844	0.957	0.008	0.918	0.879	0.967	0.014
2	Self-Attention	0.906	0.846	0.960	0.008	0.920	0.882	**0.970**	0.013
3	SS2D (Ours)	**0.915**	**0.856**	**0.962**	**0.007**	**0.926**	**0.892**	**0.970**	**0.012**

TABLE VII
PERFORMANCE EVALUATION OF BOUNDARY EXTRACTION METHODS

No.		EORSSD				ORSSD			
		$S_\alpha \uparrow$	$F_\beta^{\max} \uparrow$	$E_\xi^{\max} \uparrow$	$M \downarrow$	$S_\alpha \uparrow$	$F_\beta^{\max} \uparrow$	$E_\xi^{\max} \uparrow$	$M \downarrow$
1	HED	0.908	0.846	0.959	0.008	0.919	0.880	0.967	0.013
2	RCF	0.911	0.853	0.961	0.008	0.921	0.886	0.969	0.013
3	Sobel	0.906	0.846	0.959	0.008	0.919	0.883	0.969	0.014
4	Ours	**0.915**	**0.856**	**0.962**	**0.007**	**0.926**	**0.892**	**0.970**	**0.012**

object structure and boundary accuracy. These results indicate that SS2D is more effective for modeling spatial structural information in the proposed architecture.

3) Ablation Study on the Boundary Extraction Method: To compare the effectiveness of different boundary cues, we trained the model using boundary cues obtained from HED, RCF, Sobel, and Wavelet. For RCF and HED, edges were preextracted and used for training. As shown in Table VII, wavelet-based boundary extraction outperforms RCF and HED. This result can be attributed to the fact that RCF and HED are generic pretrained edge detectors, where false-positive edges introduce noise that interferes with identifying salient objects. Sobel also shows lower performance than wavelet, likely because Sobel relies on simple pixel intensity differences and fails to capture fine texture variations of objects.

4) Ablation Study on the SGAE Module: The SGAE module effectively injects semantic information about salient objects into low-level feature maps during feature extraction, thereby emphasizing high-level semantic cues. Fig. 10 visualizes the behavior and role of the SGAE module. The feature maps A1–A4 are enhanced based on semantic information transferred from the (A4) feature map, producing refined feature maps B1–B4. As seen in the visualization results, the refined feature maps emphasize semantic object regions more clearly compared to the original encoder feature maps. Applying the same operation to A5 also produces B5, which exhibits stronger attention around the object location.

We further conducted a comparative evaluation to determine which feature level is most beneficial for semantic information extraction. In MobileViT, the higher-level feature maps

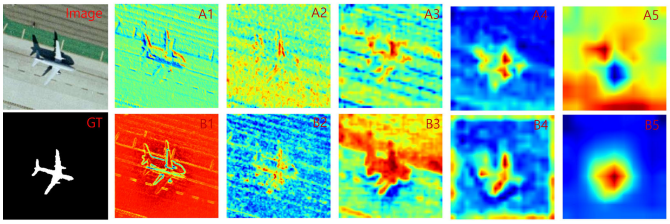


Fig. 10. Visualization of feature maps before and after refinement by the SGAE module shown in Fig. 5. (a) A1–A5 show the feature maps extracted from encoder stages 1–5, respectively. (b) B1–B5 show the corresponding refined feature maps produced by SGAE.

TABLE VIII
PERFORMANCE COMPARISON USING DIFFERENT FEATURE MAPS FOR SGAE

No.		EORSSD				ORSSD			
		$S_\alpha \uparrow$	$F_\beta^{\max} \uparrow$	$E_\xi^{\max} \uparrow$	$M \downarrow$	$S_\alpha \uparrow$	$F_\beta^{\max} \uparrow$	$E_\xi^{\max} \uparrow$	$M \downarrow$
1	E^4	0.905	0.845	0.957	0.008	0.917	0.884	0.970	0.013
2	E^5(Ours)	**0.915**	**0.856**	**0.962**	**0.007**	**0.926**	**0.892**	**0.970**	**0.012**

E^4 and E^5 contain richer semantic cues. Accordingly, we evaluated performance when semantic cues were extracted from E^4 and E^5. As shown in Table VIII, extracting semantic information from E^5 yields better performance than using E^4.

5) Ablation Study on Encoder Backbones: To demonstrate the effectiveness of using MobileViT in the WamBANet model, we conducted a comparative evaluation using different backbone encoders: 1) EfficientNet [62]; 2) ConvNeXt [63]; 3) PVT [64]; and 4) Ours (MobileViT).

As shown in Table IX, the relatively inferior performance of EfficientNet and ConvNeXt can be attributed to a structural limitation of convolution-based architectures, which have restricted receptive fields and thus fail to effectively model long-range dependencies and global context. By contrast, PVT and MobileViT achieve better generalization performance. While PVT shows comparable performance to MobileViT on some metrics, MobileViT achieves higher performance on most metrics while maintaining a lightweight architecture compared to PVT.

V. LIMITATION AND DISCUSSION

Although WamBANet is designed to enhance boundary cues under scribble-based weak supervision, several limitations remain. Since the proposed framework directly exploits high-frequency components obtained via the wavelet transform, nonsalient edges may also be unintentionally emphasized in scenes containing repetitive urban textures (e.g., roof

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TABLE IX
PERFORMANCE COMPARISON WITH DIFFERENT BACKBONE ENCODERS

No.	Encoder	Params (M) ↓	GFLOPs (G) ↓	EORSSD				ORSSD			
				$S_\alpha \uparrow$	$F_\beta^{\max} \uparrow$	$E_\xi^{\max} \uparrow$	$M \downarrow$	$S_\alpha \uparrow$	$F_\beta^{\max} \uparrow$	$E_\xi^{\max} \uparrow$	$M \downarrow$
1	EfficientNet	17.23	12.71	0.910	0.852	0.959	**0.007**	0.918	0.879	0.967	0.014
2	ConvNeXt	92.86	72.36	0.900	0.822	0.952	0.009	0.913	0.871	0.967	0.014
3	PVT	58.73	45.22	0.914	0.845	**0.962**	0.008	0.924	0.886	**0.970**	0.013
4	Ours	11.48	25.22	**0.915**	**0.856**	**0.962**	**0.007**	**0.926**	**0.892**	**0.970**	**0.012**

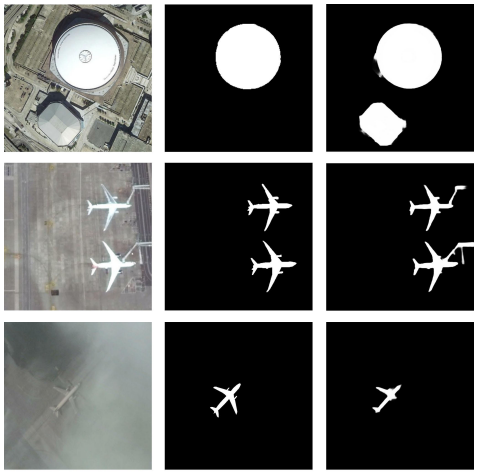


Fig. 11. Limitations of the proposed approach.

patterns or road markings) or strong sensor noise. For example, as illustrated in the first row of Fig. 11, nonsalient structures beneath the building are partially activated, while in the second row, the boarding bridge is incorrectly highlighted as a salient region. In addition, for extremely small objects or objects exhibiting low contrast against the background, the corresponding high-frequency responses may themselves be weak, thereby limiting the generation of reliable boundary cues. This issue can also be observed in the third row of Fig. 11, where the right wing of the aircraft is not clearly captured due to its low contrast with the surrounding background.

Several future research directions may address these limitations. First, adaptive frequency filtering strategies could be investigated to suppress nonsalient activations caused by repetitive textures or sensor noise, while selectively amplifying weak frequency responses associated with low-contrast objects. More advanced frequency feature refinement techniques may also help improve the robustness of structural feature extraction. Second, integrating recently proposed Mamba2- or Mamba3-based state-space modeling architectures into the WamBA module could further improve global dependency modeling and computational efficiency.

VI. CONCLUSION

This article presents WamBANet, a novel frequency-guided state-space modeling framework for scribble-based weakly supervised ORSI-SOD. A fundamental limitation of existing weakly supervised models is their reliance on computationally expensive auxiliary networks to generate boundary pseudo-labels. We address this limitation by introducing the WamBA module. By integrating the frequency-decomposition capability of DWT with the linear-time global modeling capabilities of selective SSMs, WamBANet effectively isolates and amplifies high-frequency boundary cues, maintaining structural fidelity under complex background interference. Furthermore, to resolve the inherent representational disparity between boundary-rich shallow layers and semantically dense deep layers, we propose the SGAE module. This module acts as a semantic regularizer, ensuring robust structural consistency by injecting high-level contextual cues into the low-level spatial features.

Extensive empirical evaluations on the EORSSD and ORSSD public datasets demonstrate that WamBANet outperforms previous SOTA models for weakly supervised ORSI-SOD, showing robustness across variations in object scale and background heterogeneity. In particular, the proposed architecture shows promising ORSI-SOD performance when compared with fully supervised methods. Furthermore, cross-validation testing and evaluation on camouflaged object detection demonstrate the generalization and extensibility of WamBANet.

REFERENCES

[1] K. Chen, B. Chen, C. Liu, W. Li, Z. Zou, and Z. Shi, "RSMamba: Remote sensing image classification with state space model," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.

[2] X. He, X. Han, Y. Chen, and L. Huang, "A light-weighted fusion vision mamba for multimodal remote sensing data classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 21532–21548, 2025.

[3] X. Ma, X. Zhang, and M.-O. Pun, "RS Mamba: Visual state space model for remote sensing image semantic segmentation," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.

[4] K.-B. Park, S. H. Choi, and J. Y. Lee, "M-GAN: Retinal blood vessel segmentation by balancing losses through stacked deep fully convolutional networks," *IEEE Access*, vol. 8, pp. 146308–146322, 2020.

[5] K.-B. Park and J. Y. Lee, "SwinE-Net: Hybrid deep learning approach to novel polyp segmentation using convolutional neural network and Swin transformer," *J. Comput. Des. Eng.*, vol. 9, no. 2, pp. 616–632, Apr. 2022.

[6] M. Kim, S. H. Choi, K.-B. Park, and J. Y. Lee, "A hybrid approach to industrial augmented reality using deep learning-based facility segmentation and depth prediction," *Sensors*, vol. 21, no. 1, p. 307, Jan. 2021.

[7] C. Yang, W. Wu, Y. Wang, and H. Zhou, "A novel feature-based model for zero-shot object detection with simulated attributes," *Int. J. Speech Technol.*, vol. 52, no. 6, pp. 6905–6914, Apr. 2022.

[8] H. Chen, J. Song, C. Han, J. Xia, and N. Yokoya, "ChangeMamba: Remote sensing change detection with spatiotemporal state space model," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4409720.

[9] J. Yuan, X. Zou, H. Xia, T. Liu, and F. Wu, "Bi-branch multiscale feature joint network for ORSI salient object detection in adverse weather conditions," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 2006910.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 14:26:17 UTC from IEEE Xplore. Restrictions apply.

[10] Y. Xie et al., "Localization, balance, and affinity: A stronger multifaceted collaborative salient object detector in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 4700117.
[11] Z. Huang, T.-Z. Xiang, H.-X. Chen, and H. Dai, "Scribble-based boundary-aware network for weakly supervised salient object detection in remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 191, pp. 290–301, Sep. 2022.
[12] L. Ma, X. Luo, H. Hong, Y. Zhang, L. Wang, and J. Wu, "Scribble-attention hierarchical network for weakly supervised salient object detection in optical remote sensing images," *Int. J. Speech Technol.*, vol. 53, no. 10, pp. 12999–13017, May 2023.
[13] Y. Liu, M.-M. Cheng, X. Hu, K. Wang, and X. Bai, "Richer convolutional features for edge detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 5872–5881.
[14] S. Xie and Z. Tu, "Holistically-nested edge detection," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 1395–1403.
[15] G. Xing, M. Wang, F. Wang, F. Sun, and H. Li, "Lightweight edge-aware mamba-fusion network for weakly supervised salient object detection in optical remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5631813.
[16] J. Canny, "A computational approach to edge detection," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. PAMI-8, no. 6, pp. 679–698, Nov. 1986.
[17] C. Li, R. Cong, J. Hou, S. Zhang, Y. Qian, and S. Kwong, "Nested network with two-stream pyramid for salient object detection in optical remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 11, pp. 9156–9166, Nov. 2019.
[18] Q. Zhang et al., "Dense attention fluid network for salient object detection in optical remote sensing images," *IEEE Trans. Image Process.*, vol. 30, pp. 1305–1317, 2021.
[19] Z. Tu, C. Wang, C. Li, M. Fan, H. Zhao, and B. Luo, "ORSI salient object detection via multiscale joint region and boundary model," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5607913.
[20] J. Zheng, Y. Quan, H. Zheng, Y. Wang, and X. Pan, "ORSI salient object detection via cross-scale interaction and enlarged receptive field," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.
[21] A. Gu and T. Dao, "Mamba: Linear-time sequence modeling with selective state spaces," 2023, *arXiv:2312.00752*.
[22] Y. Liu et al., "VMamba: Visual state space model," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 37, 2024, pp. 103031–103063.
[23] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 1–11.
[24] L. Gao, B. Liu, P. Fu, and M. Xu, "Adaptive spatial tokenization transformer for salient object detection in optical remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5602915.
[25] R. Yan, L. Yan, G. Geng, Y. Cao, P. Zhou, and Y. Meng, "ASNet: Adaptive semantic network based on transformer–CNN for salient object detection in optical remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5608716.
[26] Z. Huang, H. Chen, B. Liu, and Z. Wang, "Semantic-guided attention refinement network for salient object detection in optical remote sensing images," *Remote Sens.*, vol. 13, no. 11, p. 2163, May 2021.
[27] X. Wang, X. Fang, M. Jiang, J. Zhu, and Z. Li, "Edge-guided refinement network with similarity perception for salient object detection in optical remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5655014.
[28] L. Sun et al., "Local–global information perception network for salient object detection in optical remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5655014.
[29] J. Han, M. Wang, C. Zheng, Y. Hou, and F. Sun, "Spatial-frequency enhancement and scale-decoupled refinement network for salient object detection in ORSI," *IEEE Trans. Geosci. Remote Sens.*, vol. 64, 2026, Art. no. 5625115.
[30] J. He, H. Luo, X. Cai, and Z. Gao, "Cross-modal fusion with UVI enhancement for salient object detection in low-light ORSI images," *IEEE Trans. Geosci. Remote Sens.*, vol. 64, 2026, Art. no. 5628916.
[31] S. Mehta and M. Rastegari, "MobileViT: Light-weight, general-purpose, and mobile-friendly vision transformer," 2021, *arXiv:2110.02178*.
[32] L. Di, B. Zhang, and Y. Wang, "Multiscale and multidimensional weighted network for salient object detection in optical remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5625114.
[33] Y. Liu, M. Xu, T. Xiao, H. Tang, Y. Hu, and L. Nie, "Heterogeneous feature collaboration network for salient object detection in optical remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5635114.
[34] B. Zhou, A. Khosla, A. Lapedriza, A. Oliva, and A. Torralba, "Learning deep features for discriminative localization," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 2921–2929.
[35] J. Feng, H. Huang, J. Zhang, W. Dong, D. Zhang, and L. Jiao, "SA-MixNet: Structure-aware mixup and invariance learning for scribble-supervised road extraction in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5602214.
[36] A. Zhao, H. Wang, X. Li, and H. Shen, "SuperScribble: A super-pixel and pixel dual constraint remote sensing image semantic segmentation network with scribble annotation," *IEEE Trans. Geosci. Remote Sens.*, vol. 64, 2026, Art. no. 5620113.
[37] Y. Zheng, S. Liu, and L. Bruzzone, "SHIFT: Scribble-driven hybrid iterative feature tracker for multimodal remote sensing image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5636514.
[38] A. Hatamizadeh and J. Kautz, "MambaVision: A hybrid mamba-transformer vision backbone," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2025, pp. 25261–25270.
[39] J. He, K. Fu, X. Liu, and Q. Zhao, "Samba: A unified mamba-based framework for general salient object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2025, pp. 25314–25324.
[40] A. Gu, K. Goel, and C. Ré, "Efficiently modeling long sequences with structured state spaces," 2021, *arXiv:2111.00396*.
[41] A. Gu et al., "Combining recurrent, convolutional, and continuous-time models with linear state space layers," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 34, 2021, pp. 572–585.
[42] L.-C. Chen, G. Papandreou, I. Kokkinos, K. Murphy, and A. L. Yuille, "DeepLab: Semantic image segmentation with deep convolutional nets, atrous convolution, and fully connected CRFs," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 40, no. 4, pp. 834–848, Apr. 2018.
[43] D.-P. Fan, M.-M. Cheng, Y. Liu, T. Li, and A. Borji, "Structure-measure: A new way to evaluate foreground maps," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 4558–4567.
[44] R. Achanta, S. Hemami, F. Estrada, and S. Susstrunk, "Frequency-tuned salient region detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2009, pp. 1597–1604.
[45] D.-P. Fan, C. Gong, Y. Cao, B. Ren, M.-M. Cheng, and A. Borji, "Enhanced-alignment measure for binary foreground map evaluation," 2018, *arXiv:1805.10421*.
[46] F. Perazzi, P. Krähenbühl, Y. Pritch, and A. Hornung, "Saliency filters: Contrast based filtering for salient region detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2012, pp. 733–740.
[47] ptFLOPs. (2025). *FLOPs Counting Tool for Neural Networks in PyTorch Framework*. [Online]. Available: https://pypi.org/project/ptflops/
[48] G. Li, Z. Liu, Z. Bai, W. Lin, and H. Ling, "Lightweight salient object detection in optical remote sensing images via feature correlation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5617712.
[49] L. Meng et al., "Progressive enhancement of foreground features for salient object detection in optical remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 7572–7591, 2025.
[50] Y. Fu, W. Fang, and V. S. Sheng, "Rethinking masked autoencoder for salient object detection in optical remote sensing images from a pseudo image pretraining perspective," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5624212.
[51] J. Han, F. Sun, Y. Hou, J. Sun, and H. Li, "Exploring a lightweight and efficient network for salient object detection in ORSI," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5631014.
[52] J. Zhang, X. Yu, A. Li, P. Song, B. Liu, and Y. Dai, "Weakly-supervised salient object detection via scribble annotations," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 12543–12552.
[53] S. Yu, B. Zhang, J. Xiao, and E. G. Lim, "Structure-consistent weakly supervised salient object detection with local saliency coherence," in *Proc. AAAI Conf. Artif. Intell.*, Palo Alto, CA, USA, 2021, vol. 35, no. 4, pp. 3234–3242.
[54] X. Li, Y. Xu, L. Ma, Z. Huang, and H. Yuan, "Progressive attention-based feature recovery with scribble supervision for saliency detection in optical remote sensing image," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5631212.
[55] R. Yan, L. Yan, Y. Cao, G. Geng, and P. Zhou, "One-stop multiscale reconciliation attention network with scribble supervision for salient object detection in optical remote sensing images," *Int. J. Speech Technol.*, vol. 54, no. 5, pp. 3737–3755, Mar. 2024.
[56] L. Yan et al., "Multi-source information fusion attention network for weakly supervised salient object detection in optical remote sensing images," *Expert Syst. Appl.*, vol. 261, Feb. 2025, Art. no. 125505.

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[57] T.-N. Le, T. V. Nguyen, Z. Nie, M.-T. Tran, and A. Sugimoto, "Anabranch network for camouflaged object segmentation," *Comput. Vis. Image Understand.*, vol. 184, pp. 45–56, Jul. 2019.
[58] C. He et al., "Weakly-supervised concealed object segmentation with SAM-based pseudo labeling and multi-scale feature grouping," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 36, 2023, pp. 30726–30737.
[59] R. He, Q. Dong, J. Lin, and R. W. Lau, "Weakly-supervised camouflaged object detection with scribble annotations," in *Proc. AAAI Conf. Artif. Intell.*, 2023, pp. 781–789.
[60] C. He et al., "Segment concealed objects with incomplete supervision," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 47, no. 9, pp. 7832–7851, Sep. 2025.
[61] T. Han, X. Wang, Q. Dong, M. Huang, J. Jia, and F. Zhang, "Weakly supervised camouflaged object detection as progressive perception learning," *Knowl.-Based Syst.*, vol. 325, Sep. 2025, Art. no. 113993.
[62] M. Tan and Q. Le, "EfficientNet: Rethinking model scaling for convolutional neural networks," in *Proc. Int. Conf. Mach. Learn.*, 2019, pp. 6105–6114.
[63] Z. Liu, H. Mao, C.-Y. Wu, C. Feichtenhofer, T. Darrell, and S. Xie, "A ConvNet for the 2020s," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, New Orleans, LA, USA, Jun. 2022, pp. 11966–11976.
[64] W. Wang et al., "PVT v2: Improved baselines with pyramid vision transformer," *Comput. Vis. Media*, vol. 8, no. 3, pp. 415–424, Sep. 2022.



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